

Kannada Word Detection in Heterogeneous Scene Images

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Abstract— Text extraction from scene images has started gaining a lot of traction in recent years in the computer vision field as its applications is manifold. One of its sub-categories is scene text detection. Factors like complex backgrounds, curvature, orientation, image quality, and various font styles and sizes makes it a difficult task. Moreover, building a single language scene text detector in a multilingual setting, especially Indian scripts, is more challenging in contrast to a general text detector. In this work we attempt the task of Kannada text localization as well as word detection in natural scenes, which is underexplored in comparison to other well-known Indian languages. To achieve this, we fine tune a You Only Look Once (YOLOv4) based object detection model on our Kannada scene text images dataset collected. Experiments on local boards and sign images captured on smartphone, showed a mean average precision of 93.5% and per image average detection speed of 30 milli-seconds; pointing towards real time use and integration with mobile applications.

Keywords— scene text detection, kannada, word detection, neural network, deep learning

I. INTRODUCTION

Natural scene text detection and recognition is one of the most important and challenging in the computer vision domain. This involves two tasks: first, segmentation of text region in images or videos with noisy backgrounds and second, extracting the text and performing recognition to identify the words in the text.

With prominence being given to Indian languages in different states, it is a growing issue to detect and recognize text in these languages. The motivation for selecting Kannada text detection is that, we feel it's not sufficiently evaluated and also due to its real-world applications like: Aiding visually impaired persons, text transcription for tourists visiting Karnataka state, annotation tool for Kannada scene images, scene understanding for self-driving vehicles and Image-to-text or Image-to-Image retrieval when combined with text recognition.

Presence of multiple languages on boards, captured scene text images consisting of different orientations, motion blur, font size and illumination, shadows and occlusions, various font colors, styles and Ottakshara, Kagunita Akshara and Numbers support in Kannada script and ambiguity in distinguishing characters and words due to uneven spacing (unlike Hindi text for example, where all letters are connected

by a horizontal line) are some of the key challenges of Kannada text detection.

With many focusing on Optical Character Recognition (OCR) (i.e., character-level classification on scene images, handwritten and electronic materials) and locating text in image first, then applying recognition to detect a particular language text, there is a dearth of systems for word-level detection and recognition built from scratch, specifically in Kannada language.

In this work we mainly focus on detection because efficient text detection is fundamental to yield better text recognition and transliteration results, which is another popular research area.

Our aim is to build a low complexity Kannada scene text detector which can localize and detect Kannada words with high confidence in natural scenes, in the presence of English, North and South Indian language text. So, we use a lightweight YOLOv4 object detection model and modify it to perform Kannada word detection. To train the model, we furnish our own dataset as shown in Fig. 1. In addition, this method will eliminate the need to introduce a language-script identification stage in conjunction with text detection for recognition and also performing word detection using individual character segments.

The rest of the paper is organized as follows: Section II summarizes related works, Section III details the methodology used, Section IV presents the results obtained, and the paper is concluded in Section V with future scope.



Fig. 1. Sample of Kannada scene text dataset collected.

Traditional scene text detection techniques proposed use, Canny edge detection with bag of visual words, Stroke width Transform (SWT) with tree bagging or Support Vector Machines (SVM) based character or non-text region classification [1,2], surface saliency map thresholding with K-Means clustering for text segmentation and line level text recognition based on character script classification [3], Scale-Invariant Feature Transformation (SIFT) for spotting text in videos [4] and feed forward neural network with sum of zone wise vertical and horizontal features for character classification [5].

Recent developments in scene text detection can be categorized into anchor-based, region proposal networks and popular English scene text datasets like variations of ICDAR, Total-Text, MSRA and CTW are mentioned in [6]. Further, Images enhanced by CLAHE algorithm and passed to the Yolov3 model with stochastic spatial sampling pooling is used to detect scene text [7]. A synthetically generated English scene text dataset is introduced in [8], which was used to train a Fully convolutional regression network (FCRN) model and achieving a F1 score of 84.2%. Some of the other popular scene text detection methods are EAST, TextBoxes++, CRAFT[9] and DBNet++[10] which are motivated by single-shot-detectors, character-level affinity and segment map binarization with feature fusion.

For Kannada character recognition, Siddiqua et al., compare VGG19 and AlexNet with batch normalization layer by applying stochastic gradient descent optimization on the Chars74k dataset [11] to achieve an accuracy of 96% and 100% respectively in [12] and propose a AksharaNet convolutional neural network model [13] with stacked depth-wise separable convolution layers and early stopping trained on Kannada Scene Individual Character (KSIC) dataset with 46.8k images and 468 classes, outperforming MobileNetV2. In KannadaPado [14], the binarized output of Midline Analysis and Propagation of Segmentation (MAPS) algorithm on manually cropped text segment is passed to Lipi Gnani Kannada OCR for transcription with low computational complexity. In [15], a fusion of Maximally stable extremal regions (MSER) and SWT is applied to generate initial region proposals and component refinement using Generative Adversarial Network is done to localize text in multilingual scene images. A Deep Belief Network using a modified Denoising Gaussian Restricted Boltzmann Machine is proposed in [16], which achieved an accuracy of 79% on Kannada scene text character classification. Contour detection for word cropping, Histogram of Oriented Gradients feature extraction and training a word recognition model on handwritten kannada images using SVM with Radial Basis Function, achieved an accuracy of 88% [17]. A combination of Tesseract OCR on kannada boards and LIBINDIC Soundex Algorithm for transliteration is used in [18].

In addition, the FasterRCNN model with Resnet50 backbone gave an average precision of 0.98 for Urdu scene ligatures detection [19]. For Telugu language scene text detection, a DenseNet based TextBoxes architecture with K-Means clustering and non-maximum suppression trained on IndicSceneText was proposed in [20]. Apart from this, there is a work on scene text recognition of Indian languages using VGG-BiLSTM-CTC model [21], but this doesn't include Kannada.

From the exhaustive literature survey, it is evident that with many focusing on character recognition, there are not many advancements in Kannada word detection in scene images that takes the approach of object detection, though it has been attempted for other languages. In addition, there is existing work where direct text recognition is applied or text in image is detected and then script identification is applied for language specific text detection. Hence, in this work we take up the task of building a single baseline neural network for both Kannada word localization and detection. The two major steps, data collection and model training are detailed in the subsections.

A. Data Collection

Research on Kannada scene text detection lacks a foundational dataset. Multilingual datasets like ICDAR MLT 2019 [22] doesn't contain Kannada scene images and very few samples are present in the MLe2e [23] dataset. Hence, our initial step in this study was to develop a comprehensive dataset consisting of scene images with backgrounds of different complexities. We captured scene images from boards in college, local shops, road navigation and billboards from the streets of Bengaluru, at different angles with challenging backgrounds comprising of a mixture of long and short-shot images. Phone camera was used to capture these images which sometimes were unclear. Using a DSLR camera would have resulted in images with better resolution and hence the detection task would have been more accurate.

Along with the self-captured images, we collated around 100 images from the Chars74k [11] original scene images with existing Kannada word-level annotations and also manually annotated the images with character-level and without annotations in the dataset, to curate our own scene text dataset consisting of 470 Kannada word annotations in both YOLO and PASCAL VOC format.

B. Kannada text detection model

In order to perform word detection, we use the pre-trained weights from YOLOv4 [24], a state-of-the-art object detection model, already trained up to 137 convolutional layers. The reason for choosing this model is because it's fast and light, making it suitable for real time applications.

YOLOv4 is a single stage network for object detection. It consists of 3 parts, namely backbone, neck, and head. The backbone is a feature extractor, that is usually a CSPDarknet-53, a convolutional neural network architecture trained on MS COCO dataset. The backbone has five residual block modules, and the feature map outputs from the residual block modules are fused at the neck of the YOLO v4 network. Next, the neck consists of the Spatial Pyramid Pooling module and Path Aggregation network. Finally, the head, YOLOv3 in this case, processes all the features aggregated and predicts bounding boxes, and it makes use of anchor boxes to detect classes of objects in an image.

As we didn't have access to a large dataset, transfer learning is employed, wherein another model trained to perform a similar task acts as a feature extractor, instead of training the model from scratch. Here the neural network treats Kannada words as objects and tries to localize text regions and predicts the confidence score for each text segment or coordinates identified to be Kannada.

The word detection pipeline, which involves training and testing is shown in Fig. 2. The entire data is split in 90:10 ratio, for obtaining training and testing images. The model trains to reduce CioU (Intersection over union) loss or the bounding box regression loss and calculates mean-average-precision (mAP) periodically to monitor performance and trains until maximum number of iterations is reached or there is no major change in the average loss.

The default configuration file for the model is modified as highlighted in Table II, where [net] represents the configurations to be leveraged during custom model training and testing. Here, 'classes' = 1 and 'max_batches' = (classes * 2000) and steps is 80% and 90% of 'max_batches'. The 'filters' parameter in [convolution] layer before each of the 3 [yolo] layers is set using (classes + 5) x 3 and 'random = 1' in the final [yolo] layer is used to perform multi-resolution training, where there is +-50% shift from the base resolution of the images every 10 batches, this is responsible for improving model learning with a small dataset. The model is trained using Google Colaboratory with GPU for faster computation. The custom trained weights has 162 layers.

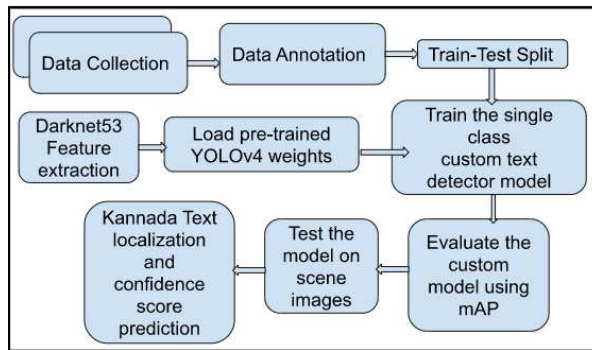


Fig. 2. Kannada word detection pipeline.



Fig. 3. Bounding box and confidence score on complex test images.



Fig. 4. Bounding box and confidence score on sample test images.

TABLE I. QUANTITATIVE RESULTS OBTAINED ON TEST DATA

Parameter	Value	Parameter	Value
No. of classes	1	Class Name	Kannada
True Positives	83	False Positives	22
False Negatives	3	IoU Threshold	0.50
Precision	0.79	Recall	0.97
F1 Score	0.87	Mean Average Precision@0.5	0.935

IV. EXPERIMENTAL RESULTS

To measure performance of the custom text detection model, the metric calculated is mAP score with 0.5 IoU threshold. Here, IoU represents Intersection over Union which determines how close the prediction is to the ground truth bounding box coordinates. Precision and recall is calculated based on the predicted IoU score (determines true, false positives and negatives) for a pre-defined IoU threshold. Average Precision is calculated by finding the area under the precision-recall curve and as the number of classes is 1 in our case, it is equal to the mean average precision computed. Localization and Detections of Kannada words by our model on diverse test images are shown in Fig. 3. and Fig. 4. The best custom weights saved after 1000 iterations, when applied on the test set showed an average precision score of 93.48% as shown in Table 1. Greedy non maximum suppression is applied to filter bounding boxes below detection score of 0.3. To avoid bias, model inference was performed on some images not in test or train set as shown in Fig. 6. And Fig. 7. Furthermore, considering other factors such as number of bounding boxes correctly predicted, word bounding box coordinates and the confidence score for class label=kannada, it can be said that the model is holistically performing well. From multiple experiments it was noted that the mean prediction time was approximately 30 milli-seconds per image. So, this model is suitable for performing detections on live video feed as well. The proposed model is efficiently able to distinguish between other languages and Kannada and this has been achieved while training the model on less data. However, training the model on more data will yield much better results in terms of word segmentation to aid in efficient text recognition.

V. CONCLUSION AND FUTURE WORK

In this paper, an object detection approach for Kannada scene text detection has been proposed on scene image dataset collected from the streets of Bengaluru. Detection has been done at word level on scene images and the class has been assigned to the detected region using the Yolov4 algorithm. It was observed that our algorithm was able to draw bounding boxes and predict the class with a mean average precision of 93.48 % using the final custom trained weights at 50% IoU threshold and 30 milli-seconds mean prediction time. It was also noted that although our model was able to detect and classify Kannada words accurately in varied backgrounds, the model struggled to localize Kannada with high precision in extreme angles and orientations and sometimes detected other language texts but with comparatively less confidence. This can be addressed by increasing diversity in the dataset or by replacing the model architecture. Moreover, the hyper parameter, IoU threshold can be tuned to measure model performance and improve the confidence scores of the detections.

This study has been limited to detection of Kannada language at a word level. It can be further taken forward to build an end-to-end detection and recognition network with the help of OCR modules. The proposed model also supports applications like image-based Kannada dictionary and language translation to mainly aid tourists, blind persons, and autonomous vehicles, when extended to perform detections on videos as Yolov4 is known for real time usage.

Although Kannada Scene text detection has been doing the rounds for some time, it is still in its nascent stages. The future scope of this work would be to first improve the detection model by enhancing our dataset along with synthetically generated scene text dataset with word annotations and then performing word recognition to develop a robust comprehensive model for Kannada language. And finally, to integrate it with a mobile application to implement any of the above-mentioned applications.



Fig. 5. Sample detection results by the model.



Fig. 6. Sample detection results by the model.

TABLE II. MODIFICATIONS (BOLD TEXT) IN DEFAULT CONFIGURATION FILE

Custom YOLOv4 model configuration ^a
[net] Testing batch=1 subdivisions=1 Training batch=64 subdivisions=16 width=416 height=416 channels=3 momentum=0.949 decay=0.0005 angle=0 saturation = 1.5 exposure = 1.5 hue=.1 learning_rate=0.001 burn_in=1000 max_batches = 2000 policy=steps steps=1600,1800 scales=.1,.1

^a. Modified from <https://github.com/AlexeyAB/darknet>

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